Materials Science & Engineering A 953 (2026) 149792

Contents lists available at ScienceDirect

Materials Science & Engineering A

journal homepage: www.elsevier.com/locate/msea

Duplex microstructure enhanced mechanical property and underlined mechanism in Fe35MnxAl0.1C low-density steel

Qingfeng Kang [a,b], Zexi Zhang [a,b], Hui Wang [a,b], Haifeng Xu [a,b], Cunyu Wang [a,b], Zhengdong Liu [a,b], Wenquan Cao [a,b,*]

[a] Special Steel Department, Central Iron and Steel Research Institute (CISRI), Beijing, 100081, China
[b] State Key Laboratory of Advanced Special Steel, Beijing, 100081, China

ARTICLE INFO

Keywords:
Low-density steel
Mechanical properties
Edge dislocation
Slip band
DO_3 ordered domains

ABSTRACT

The microstructure and mechanical properties of Fe-35Mn-xAl-0.1C (x = 4, 6, 8, 10) low-density steels were studied using scanning electron microscopy (SEM), transmission electron microscopy (TEM) and tensile machine. It is found that the microstructure evolves with the increasing of Al content from a single austenite phase steel for 4 %Al, to a ferrite-based duplex steel for 8 %Al, but nearly a fully ferrite phase steel with 10 %Al. As the Al content is increased from 4 wt % to 8 wt %, the yield strength and tensile strength are significantly increased from 239 MPa to 552 MPa–455 MPa and 740 MPa, respectively, while the uniform elongation is only slightly decreased from 38.5 % to 31.0 %. However, as the Al content is increased up to 10 %, the yield strength is gradually increased to 585 MPa, but the tensile elongation is sharply decreased down to about 4.6 %. Based on the thorough microstructure examination of the dislocation type and dislocation pattern and the microstructure evolution during tensile deformation, it is concluded that the excellent combination of the mechanical property of the 8Al steel is realized by the activation of the edge-typed dislocation and dynamic slip band refinement in the ferrite grain mitigated by the surrounded austenite grains, which enhanced the strain hardening rate significantly. This study not only reveals a new ductility-enhancing mechanism of ferrite + austenite duplex steel, but also provides a new route for designing BCC based low-density steel with high strength and high ductility.

1. Introduction

As one of the new generation materials, the lightweight steel has been paid much more attention over the past two decades due to its excellent combination of mechanical properties and low density [1–6]. The lightweight steels mainly include austenitic low-density steel with high carbon, high manganese, and high aluminum contents, the duplex low-density steel with relatively low contents of carbon, manganese, and aluminum, and the ferritic low-density steel with the lowest contents of carbon, manganese, and aluminum [2]. Al-alloyed low-density steels exhibit an attractive combination of yield strength, tensile strength, elongation and work-hardening capacity [2,7–9]. The excellent mechanical properties of high stacking fault energy (SFE) austenitic low-density steel are attributed to the planar glide of dislocations and the dynamic slip band refinement (DSBR) mechanism [10–13]. The planar glide of dislocations in high stacking fault energy austenite originates from the “slip plane softening” effect caused by dislocation shearing of ordered domains [12].

In ferritic low-density steel, plastic instability exists and is induced by the introduction of B2 and DO_3 ordered domains [14–16]. It was found that the dislocations gradually transitioned from single mixed dislocations to paired screw dislocations with the increase of Al content in Fe-Al ferritic low-density steel [14,16], which was associated with low-energy screw dislocations [14] and high Peierls-Nabarro stresses [16]. In traditional fully body-centered cubic (BCC) alloys, the gliding velocities of edge dislocations are typically much higher than those of screw dislocations due to the low Peierls-Nabarro stresses [17–19]. As a result, edge dislocation segments are readily annihilated during deformation, and screw dislocations survive in the deformed microstructure [20,21]. However, edge dislocations are observed in multi-component BCC phases, such as titanium alloys [22] and high-entropy alloys [20,21]. In these alloys, the significant increase of Peierls-Nabarro stress for edge dislocation glide, caused by substantial local atomic volume mismatch or ordered domains, may effectively impede dislocation

* Corresponding author. Special Steel Department, Central Iron and Steel Research Institute (CISRI), Beijing, 100081, China.
E-mail address: caowenquan@nercast.com (W. Cao).

https://doi.org/10.1016/j.msea.2026.149792
Received 13 October 2025; Received in revised form 16 December 2025; Accepted 14 January 2026
Available online 15 January 2026
0921-5093/© 2026 Published by Elsevier B.V.

motion and consequently promote extensive dislocation multiplication [20]. However, edge dislocations in ordered ferrite have not been reported to date in ferritic matrix lightweight steels.

The research on duplex low-density steels primarily focused on austenite-based duplex low-density steels [23–25]. The introduction of B2 and DO_3 ordered phases into high-aluminum ferrite can bring a significant enhancement in yield strength, but simultaneously lead to an increase in brittleness [15,23,24,26,27]. It was found that B2 and DO3 ordered domains appear in Fe-Al alloys during the cooling process when the Al content exceeds 6.2 % [28,29]. In the Fe21Mn10Al1C5Ni lightweight steel, dislocations within the B2 particles exhibit planar glide characteristics, which may be attributed to the dislocation shearing of DO_3 nanoparticles [24]. Furthermore, planar slip of dislocations associated with the shearing of ordered structures has also been observed in titanium alloys [30,31] and high-entropy alloys [32]. Therefore, the dislocation deformation characteristics of ferrite will be significantly influenced by the interactions between dislocations and ordered phases. However, the systematic work in this aspect has not yet been conducted to investigate the work hardening mechanisms of high-aluminum ferritic matrix-based lightweight steels.

To address this issue of both dislocation patterns and work hardening behaviors, Fe35MnxAl0.1C (x = 4, 6, 8, 10 wt%) low-density steels were designed. Among them, Fe35Mn4Al0.1C is an austenitic lightweight steel, Fe35Mn6Al0.1C is an austenite based duplex lightweight steel, and Fe35Mn8Al0.1C and Fe35Mn10Al0.1C are ferrite based duplex lightweight steels. Tensile deformations were carried out to reveal the microstructural effects on the work hardening behaviors and mechanical properties of the designed steels. And both scanning electron microscope (SEM) and transmission electron microscope (TEM) were applied to reveal the microstructure evolution in terms of dislocation patterns of the designed steel during deformation process. The effects of microstructure on the dislocation type, dislocation pattern, work hardening behavior and mechanical property will be discussed, which is expected to give a thorough understanding on the evolution of dislocation pattern and the work hardening behavior of the ferrite-based duplex lightweight steel.

2. Materials and experimental details

Four ingots with nominal compositions of Fe-35Mn-xAl-1C-0.1Nb-0.1V-1Mo-1W (x = 4, 6, 8, 10 wt%) were prepared via vacuum induction melting under a high-purity Ar atmosphere. The as-cast ingots were homogenized at 1150 °C for 4 h, and subsequently hot-forged into rods with a diameter of 16.5 mm. The bars were solution-treated at 1050 °C for 2 h, followed by oil quenching, to achieve a microstructure with nearly equiaxed grains. For simplicity, the four solution-treated steels are hereafter referred to as 4Al, 6Al, 8Al, and 10Al steels, respectively. The compositions of the solution-treated samples were examined by the wet-chemical analysis, and the results were shown in Table 1. The compositions of austenite phase and ferrite phase of the designed steels were investigated using the electron probe microanalysis (EPMA). The carbon content in austenite was calculated using the chemical composition lever rule [1,24]. The densities of the investigated steels, measured using the principle of Archimedes, are 7.42, 7.18, 6.99, and 6.84 g cm^{-3}, respectively.

The stacking fault energy (γ_{SFE}) of the austenite for the investigated steels was evaluated based on the well-accepted thermodynamic approach [33,34]:

$$\gamma_{SFE} = 2\rho\Delta G^{\gamma\to\varepsilon} + 2\sigma \quad (3)$$

where ρ is the molar surface density along {111} planes, $\Delta G^{\gamma\to\varepsilon}$ is the molar Gibbs energy of the phase transformation from γ to ε, and σ is the interfacial energy per unit area of the γ/ε phase boundary. The calculated γ_{SFE} was recorded in Table 2 by using thermodynamic functions and parameters listed in literature [35].

The microstructure was characterized in details by electron backscatter diffraction (EBSD) in a scanning electron microscope (JSM-7800 F) and transmission electron microscopes (TEMs) operated at 200 kV (FEI-Tecnai G2 F20 and FEI talos F200X). The original data obtained from EBSD with a scanning step size of 1 μm was analyzed by the HKL channel 5 software. Tensile testing was performed on standard specimens (5 mm in diameter, 25 mm in gauge length) machined from water-quenched steel rods, with the tensile axis oriented parallel to the rod axis. The stress-strain responses and corresponding strain hardening rates of solution-treated specimens were determined with a strain rate of 1.0×10^{-3} s^{-1} at room temperature using an MTS tensile machine. At least three tensile specimens were tested to fracture to ensure experimental reproducibility. Interrupted tensile tests were conducted to engineering strains of 3 %, 10 %, 20 %, and 28 % to investigate the microstructural evolution and to elucidate the deformation mechanisms.

Table 1
Chemical compositions of the investigated steels.

	Element concentration (wt.%)							
	Mn	Al	C	Nb	V	Mo	W	Fe
4Al	33.92	4.13	0.094	0.11	0.11	1.20	0.98	Bal.
6Al	35.14	5.97	0.095	0.11	0.10	1.24	0.95	Bal.
$6Al_\gamma$	35.92	5.88	0.098	0.09	0.09	1.12	0.86	Bal.
$6Al_\alpha$	32.44	6.83	–	0.09	0.18	1.54	1.63	Bal.
8Al	34.56	8.10	0.076	0.11	0.11	1.31	1.27	Bal.
$8Al_\gamma$	36.06	7.10	0.253	0.08	0.10	0.84	0.89	Bal.
$8Al_\alpha$	33.84	8.26	–	0.08	0.11	1.25	1.25	Bal.
10Al	34.66	10.17	0.097	0.11	0.11	1.30	1.38	Bal.
$10Al_\gamma$	37.39	8.51	–	0.08	0.09	0.80	0.81	Bal.
$10Al_\alpha$	34.13	10.18	–	0.08	0.11	1.24	1.26	Bal.

3. Results

3.1. Initial microstructures

The microstructures of the 4Al, 6Al, 8Al, and 10Al steels after solution treatment are shown by the phase distribution maps in Fig. 1(a–d). EBSD analysis indicates that 4Al steel is a single-phase face-centered cubic (FCC) γ austenite lightweight steel, 6Al steel is an austenite-based duplex lightweight steel, and 8Al and 10Al steels are ferrite-based duplex lightweight steels. It also shows that the microstructures with quasi-equiaxed grains are obtained under the solution treatment conditions. It can be observed that the austenite grains contain numerous annealing twin boundaries (represented by white lines).

Fig. 1(e–h) shows the selected area diffraction patterns (SADPs) of 4Al, 6Al, 8Al, and 10Al steels along the $[100]_\gamma$ zone axis of the austenite grains, where obvious satellite diffraction spots can be seen near the γ diffraction spots. The satellite diffraction spots near the γ diffraction spots are related to the concentration fluctuations of Al and C atoms [11,36], indicating that spinodal decomposition occurred during quenching. Very weak diffraction spots are identified at the $\{110\}_\gamma$ positions in 6Al, 8Al, and 10Al steels. In previous studies, this kind of ordered structure is revealed to be an $L1_2$ type short-range ordered crystal structure [11,37]. It can be seen that the intensity of these satellite diffraction spots decreases with increasing of the nominal Al content, which is revealed to be an $L1_2$ type short-range ordered crystal structure.

Fig. 1(i–k, m) shows the selected area diffraction patterns (SADPs) of 6Al, 8Al and 10Al steels along the $[110]_\alpha$ zone axis of the ferrite grains, where distinct superlattice diffraction spots are observed. When h, k and l (Miller indices) are all even, the diffraction spots belong to both B2 and

Table 2
γ_{SFE} at room temperature.

Steel	$4Al_\gamma$	$6Al_\gamma$	$8Al_\gamma$	$10Al_\gamma$
γ_{SFE} (23 °C)/mJ·m^{-2}	51.25	62.64	68.47	–

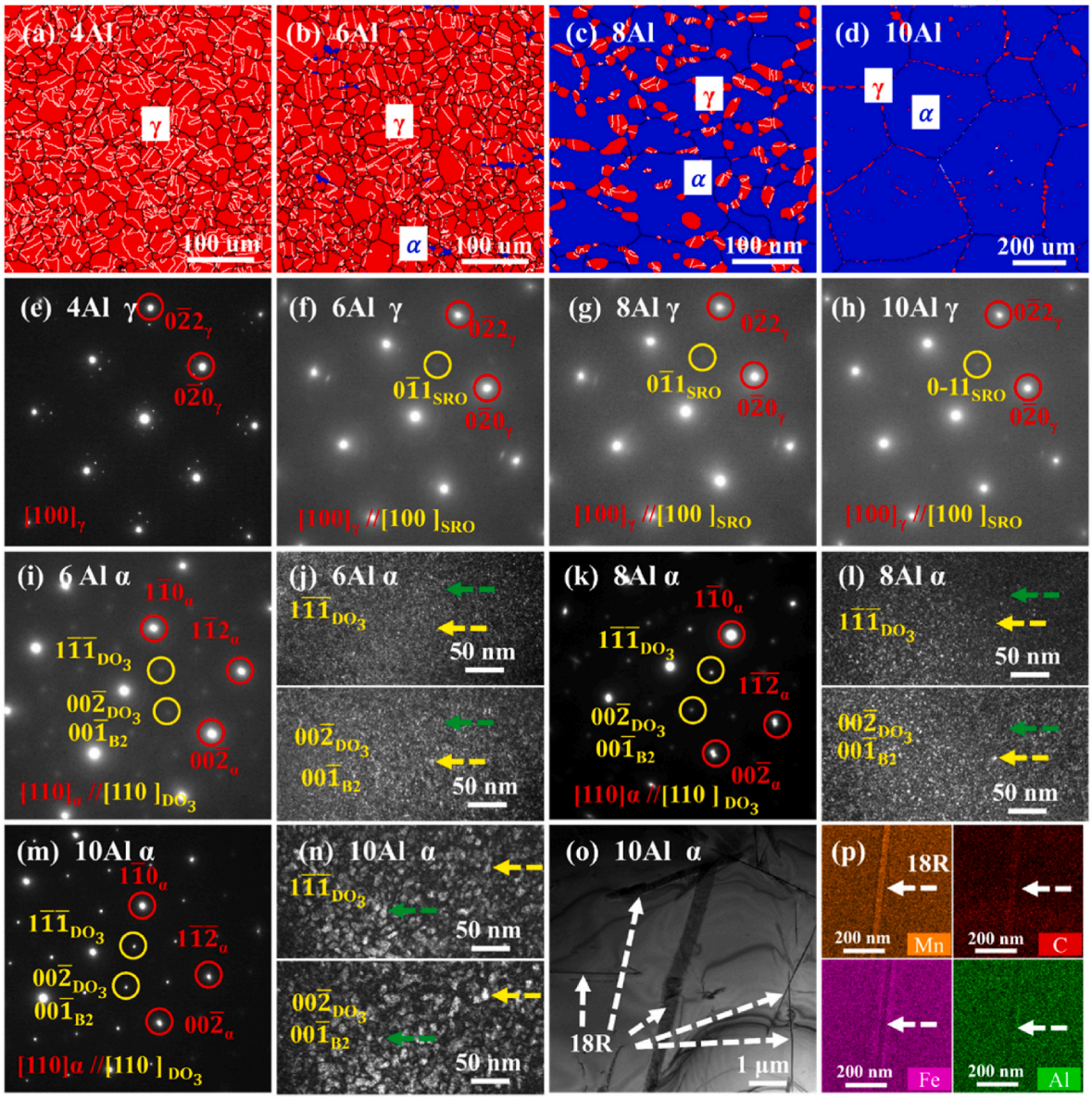

Fig. 1. Microstructures of the (a, e) 4Al, (b, f, i, j) 6Al, (c, g, k, l) 8Al and (d, h, m, n) 10 Al steels. (a–d) EBSD phase map, where red represents austenite and blue represents ferrite. (e–h) Selected area diffraction patterns along the $[001]_\gamma$ zone axis. (i, k, m) Selected area diffraction patterns along the $[110]_\alpha$ zone axis, where the primary α diffraction spots and superlattice reflections from the DO_3 are shown. (j, l, n) TEM dark field images using the (1 $\bar{1}\bar{1}$) and (00 $\bar{2}$) DO_3 superlattice reflection. (o) TEM morphology of 18R martensite. (p) EDS mappings of 18R martensite. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

DO_3. When h, k and l are all odd, the diffraction spots belong only to DO_3 [11,16]. Fig. 1(j–l, n) presents the DO_3 dark-field (DF) images acquired from the $\{111\}_{DO3}$ and $\{002\}_{DO3}$ diffraction spots in the same field of view for the 6Al, 8Al and 10Al steels, respectively. The green arrows show that the dark-field image acquired from the $\{002\}_{DO3}$ reflection exhibits a lower intensity or approximate intensity than that obtained from the $\{111\}_{DO3}$ reflection, indicating the coexistence of main DO_3 domains in the ferritic matrix. The yellow arrows indicate that the dark-field image acquired from the $\{002\}_{DO3}$ reflection exhibits a higher intensity than that obtained from the $\{111\}_{DO3}$ reflection, indicating the coexistence of DO_3 domains and B2 domains within the ferritic matrix. Therefore, DO_3 ordered domains coexist within both the B2 ordered domains and the ferrite matrix [23]. The maximum sizes of DO_3 ordered domains in the 6Al, 8Al, and 10Al steels are estimated to be approximately 2.0 nm, 2.1 nm, and 19 nm, respectively. The size of the DO_3 ordered domains increases with the increase of the nominal aluminum content, suggesting that Al promotes the precipitation of DO_3 ordered domains. In addition, acicular precipitates are observed within the ferrite grains of the 10Al steel, as shown in Fig. 1(o and p). It can be observed that Mn and C are significantly enriched in the acicular precipitates, while Al is depleted. The precipitates are identified as 18R martensite formed within the ferrite of the Fe-Mn-Al-C alloy during quenching [38–41]. This type of acicular martensite can significantly reduce the ductility of ferrite [42]. The TEM EDS mapping analysis indicates that the 18R martensite is rich of manganese and carbon, as displayed in Fig. 1(p).

X-ray diffraction (XRD) analysis was conducted to further examine the phase composition and mass fractions in the lightweight steels, as presented in Fig. 2(a). The results indicate that the mass fractions of ferrite are 75.7 % and 98.1 % for 8Al and 10Al steels, respectively. The microhardness of austenite and ferrite in the designed steels is shown in Fig. 2(b). It can be seen that the microhardness of ferrite is significantly

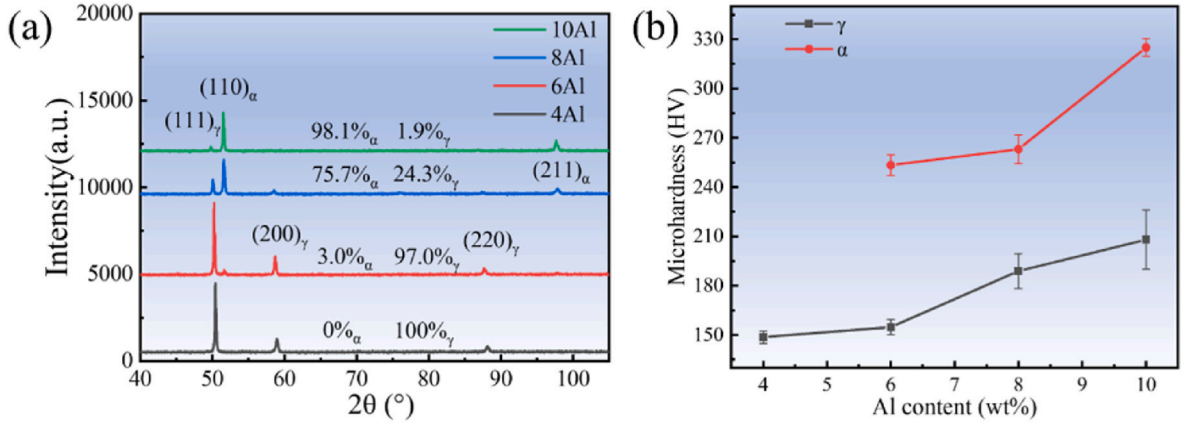

Fig. 2. XRD profiles and microhardness of 4Al, 6Al, 8Al, and 10Al steels. (a) XRD profiles to show the aluminum content effects on phase fraction. (b) Microhardness affected by aluminum contents.

higher than that of austenite, indicating that ferrite acts as a strengthening phase in the designed steels.

Based on above analysis on the microstructure of the low-density steels with different Al content as shown in Fig. 1, it can be concluded that the microstructure of the low-density steel is strongly affected by the aluminum content, giving nearly full austenite microstructure for 4 %Al and 6 %Al,ferrite based duplex microstructure for 8 %Al, but nearly full ferrite microstructure for 10 % AL. Furthermore, in the austenite phase, the short-range ordering of Al and C could be found in the austenite phase of the low-density steel with aluminum evolved from 6 % to 10 %. In the ferrite phase, both B2 and DO_3 could be found in the ferrite phase, indicating that both ordered phases are easily developed in the ferrite phase.

3.2. Mechanical properties

The typical tensile engineering stress-strain curves and mechanical properties of 4Al, 6Al, 8Al, and 10Al steels are shown in Table 3 and Fig. 3(a and b). It can be observed that as the Al content increases from 4 wt% to 8 wt%, the yield strength and tensile strength increase from 239 MPa to 552 MPa–455 MPa and 740 MPa, respectively, while the uniform elongation only decreases from 38 % to 31 %. However, as the Al content increased up to 10 %, the yield strength is increased to 585 MPa, but the tensile elongation is only about 4.6 %, exhibiting the brittle characteristics. As demonstrated in Section 3.1, the degradation in plasticity of the 10Al steel can be attributed to the coarsening of DO_3 ordered domains [15] and the formation of 18R martensite [42].

Fig. 3 (c, d) presents the true stress-strain curves and the corresponding strain hardening curves for the 4Al, 6Al, 8Al, and 10Al steels. It can be seen that the 4Al and 6Al steels exhibit typical three-stage strain hardening characteristics, labeled as A, D and E, whereas the 8Al steel displays five distinct strain hardening stages, labeled as A, B, C, D and E, which is similar to the strain hardening behavior exhibited by high-carbon, high-aluminum austenitic lightweight steels. This indicates that the deformation mechanism of the designed 8Al ferrite-based duplex lightweight steels differs from that of conventional ferrite-based duplex steels [43,44]. Fig. 3 (e) presents the evolution of microhardness with strain for the 4Al and 8Al steels it can be observed that the microhardness of austenite in both steels increases with increasing strain. However, the microhardness of ferrite in the 8Al steel initially decreases and then increases with strain. That is, the ferrite in the 8Al steel exhibits softening in the early stages of deformation, indicating differences in the deformation mechanisms between ferrite and austenite in the 8Al steel.

Table 3
Mechanical properties of the investigated steels.

Test temperature	Yield strength, MPa	Ultimate tensile strength, MPa	Uniform elongation, %	Total elongation, %
4Al	239 (±7)	552 (±12)	38.5 (±2.0)	52.0 (±1.8)
6Al	251 (±3)	589 (±18)	37.6 (±0.9)	51.6 (±1.5)
8Al	455 (±10)	740 (±14)	31.0 (±1.3)	39.8 (±1.8)
10Al	585 (±29)	613 (±41)	2.1 (±1.1)	4.6 (±1.6)

3.3. Deformed microstructure characteristics

The microstructure of the uniform deformation zone of the experimental steels after tensile deformation is shown in Fig. 4. Due to the plastic instability of the 10Al steel, the microstructures after fracture at ~3 % engineering strain are shown in Fig. 4(g–i). It can be observed that the deformation mechanisms of austenite (Fig. 4(a, b, d, e, g)) and ferrite (Fig. 4(c–f, h, i)) are distinctly different. The microstructure of the austenite after tensile deformation is characterized by dislocation cell blocks regardless of the aluminum content, as shown in Fig. 4(a, b, d), which is common in low-carbon or low-aluminum lightweight steels [1, 45,46]. Additionally, a small number of deformation twins are observed in 8Al steel (Fig. 4(e)), which is associated with the localization of stress [1,47]. Fig. 4 (g) shows that the 10Al steel exhibits complex planar slip characteristics during the early deformation stage of austenite, which is similar to the deformation structure of the 6Al steel reported in our previous study [1]. The difference in dislocation deformation structure between this kind of low carbon austenitic lightweight steel and the traditional high carbon austenitic lightweight steel may be associated with the carbon induced suppression of cross-slip [46] and the development of the ordered structure [11].

It is very interesting that the dislocations in ferrite exhibit distinct planar slip characteristics, as shown in Fig. 4(c–f, h, i), which is quite rare in BCC structures [20,21]. It can be observed that the ferrite dislocation structure in 8Al steel exhibits a more pronounced planar slip characteristic compared with that in 6Al steel, indicating that the increase in Al content may promote planar slip of ferrite dislocations. However, the increase in Al content also makes the 10Al steel particularly brittle. Fig. 4(h, i) shows that only a small number of long straight dislocations are activated in the ferrite matrix. These long straight dislocations are identified as screw dislocations based on the discussion in Section 4.1. Brittle fracture caused by the activation of only a few long straight screw dislocations has also been observed in the TiZrHfNb0.4 BCC high-entropy alloy [20]. The planar slip of dislocations in ferrite will be discussed in detail in Section 4.2.

To further understand the deformation mechanism of this ferrite-based lightweight steel, the microstructures of 8Al steel at different

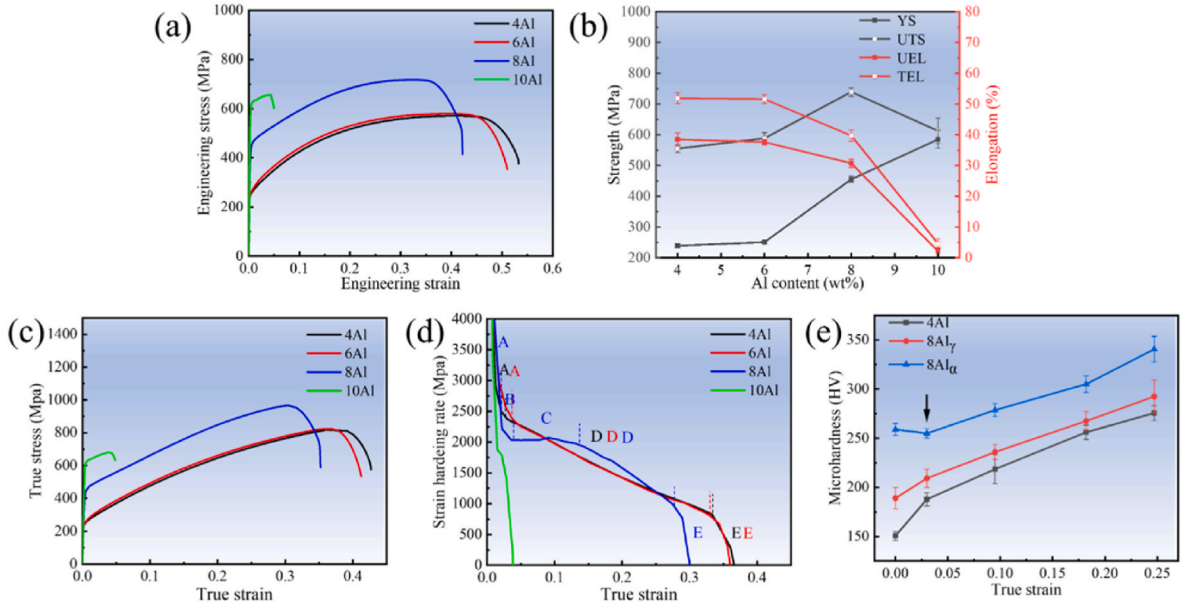

Fig. 3. Mechanical properties of 4Al, 6Al, 8Al, and 10Al steels. (a) Engineering stress-strain curves. (b) Mechanical properties. (c) True stress-strain curves. (d) Strain hardening rate-strain curves. (e) Microhardness.

engineering strains are characterized, as shown in Fig. 5.

When the engineering strain reaches 3 %, planar dislocation slip along the slip planes and distinct cross-slip are observed in the austenite (Fig. 5 (a)), while the ferrite is primarily characterized by a planar dislocation structure (Fig. 5 (b)). Fig. 5(c) presents a magnified view of the region marked in Fig. 5(b), where it can be observed that edge dislocations, mixed dislocations, and long straight screw dislocations are activated in ferrite of 8Al steel and form slip bands, which is different from the local screw dislocations in ferrite of 10Al steel. As the strain further increases to 10 %, cross-slip of dislocations in austenite becomes more pronounced (Fig. 5(d)), and the slip bands composed of dislocations in ferrite continue to evolve (Fig. 5(e)). When the engineering strain further increases to 20 %, the dislocation cell blocks (Fig. 5(f)) and dislocation cells (Fig. 5(g)) are formed in the austenite, and the slip bands in the ferrite are further refined (Fig. 5(h)). When the engineering strain reaches 28 %, the dislocation cell blocks (Fig. 5(i)) and dislocation cells (Fig. 5(j)) in the austenite are further refined, and a small number of deformation twins are formed due to the localization of stress (Fig. 5 (k)) [1]. Fig. 5(l) displays the dislocation structure in ferrite at 28 % engineering strain, revealing continued refinement of slip bands along with distinct cross-slip of dislocations. The orientation dependence of dislocation cell blocks and dislocation cell substructures in austenite has been analyzed in detail in the literature [1,45,46]. The average slip band (SB) spacing, dislocation cell block (DCB) size, and dislocation cell (DC) size in austenite, as well as the average slip band spacing in ferrite at different deformation stages, are statistically analyzed, as shown in Fig. 6.

4. Discussion

4.1. Identification of dislocation types in ferrite of 8Al and 10Al steel

There are significant differences in the micro-deformation behavior in ferrite of 10Al steel and 8Al steel, which are closely related to the activation of different types of dislocations. Here, the two-beam diffraction mode has been employed to investigate the types of dislocations in ferrite of 10Al steel and 8Al steel. To better observe the dislocations, the samples are pre-deformed to the 3 % engineering strain. Fig. 7(a–f) presents the dislocation structures in ferrite of 10Al steel using the different g diffraction vectors near the [111] zone axis.

To identify the Burgers vector b of the ferrite, the $g{\cdot}b$ analyses are conducted. When $g = [10\,\bar{1}]$ and $g = [1\,\bar{1}\,0]$ operating vector, some long straight dislocations systematically lose contrast (Fig. 7(a and b)), whereas when $g = [0\,\bar{1}\,1]$ operating vector, all dislocations are visible (Fig. 7(c)). It can be concluded that the Burgers vector b is determined to be $a/2[111]$ by the $g{\cdot}b = 0$ analysis of the ferrite in 10Al steel, as shown in Fig. 7(a–c). Similarly, the Burgers vector of dislocations in ferrite of 8Al steel is also analyzed near the [111] zone axis, as shown in Fig. 7 (g–l). Some dislocations systematically lost contrast under the $[0\,\bar{1}\,1]$ and $[\bar{1}\,10]$ operating vector conditions. Subsequently, the Burgers vector $\boldsymbol{b}$ is also determined to be $a/2[111]$ through the $g{\cdot}b = 0$ analysis of the ferrite in 8Al steel.

Fig. 7(a–l) shows that the dislocation structures are quite different although the Burgers vectors of dislocations in ferrite of both 10Al steel and 8Al steel are of the $a/2 < 111 >$ type. A predominant fraction of the dislocations in the ferrite of 10Al steel are identified as long straight screw dislocations, and their distribution is non-uniform, as shown in Fig. 7(a–f). These long straight screw dislocations are commonly observed in Fe-Al ordered alloys [14–16,48], which can be attributed at least partially to the significant energy difference between edge and screw dislocations, where the energy of screw dislocations is notably lower than that of edge dislocations [14].

Comparing with the long straight screw dislocations in 10Al steel, a substantial number of edge dislocations and mixed dislocations with near-edge character are uniformly distributed in the ferrite phase of 8Al steel. This homogeneous distribution of dislocations can effectively mitigate brittle fracture caused by stress concentration. It is reported in conventional body-centered cubic (BCC) alloys, the gliding velocities of edge dislocations are typically much higher than those of screw dislocations [18,19], owing to the lower resistance stress. Consequently, the edge segments are more readily annihilated within BCC alloys, leaving screw dislocations as the predominant type in the deformed microstructures [21]. However, edge dislocations have also been reported in the BCC-structured Mo-Nb-Ti multi-principal element alloy (MPEA) [49] and Ti-Zr-Hf-Nb high-entropy alloy (HEA) [20]. The local chemical fluctuations in these alloys result in relatively high glide resistance for edge dislocations on the glide plane, making it less likely for edge dislocations to annihilate. Similarly, B2 and DO_3 ordered domains increase the resistance stress to the glide of edge dislocations in the ferrite of 8Al steel, which reduces the glide velocity of the edge dislocations and

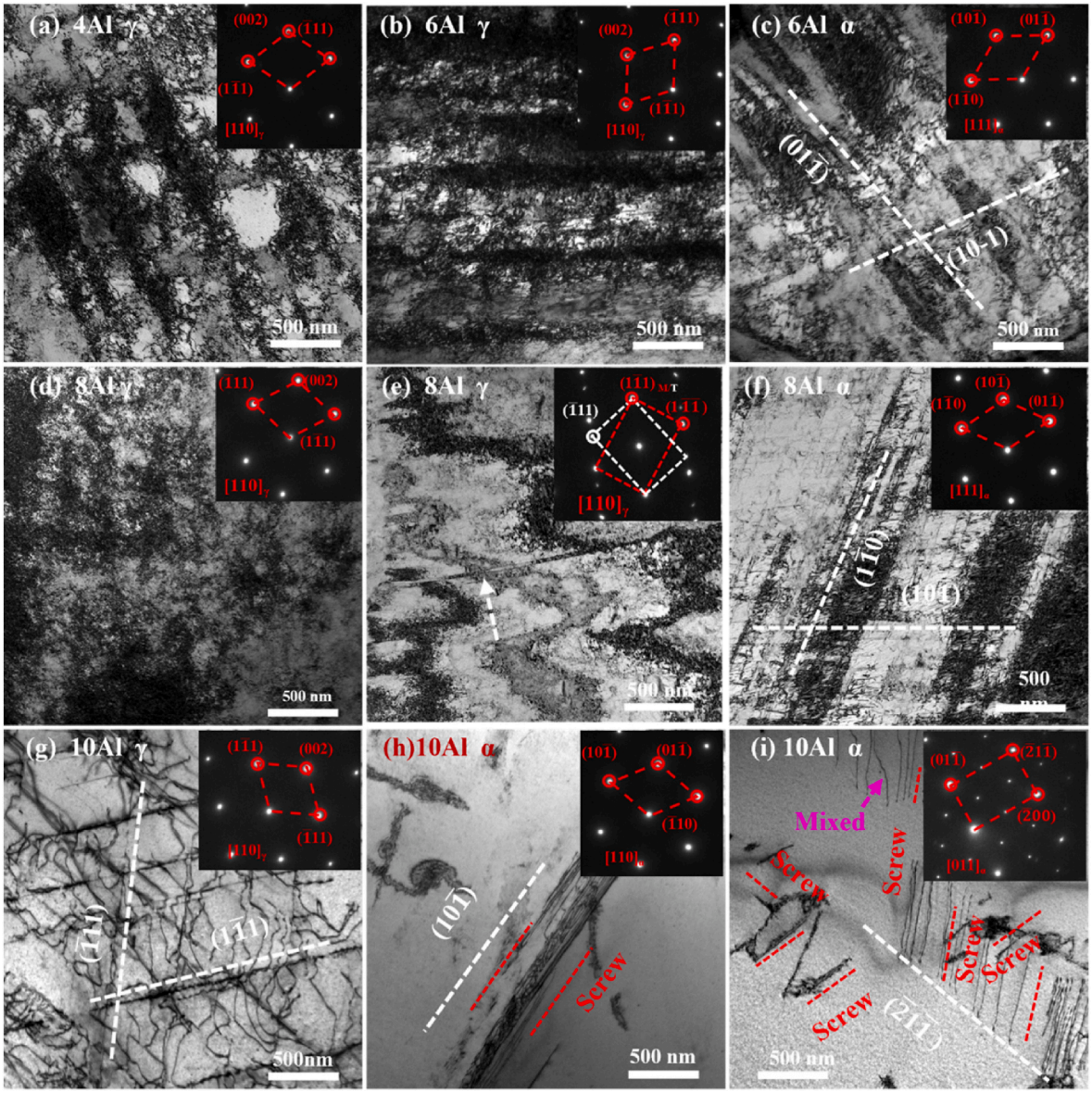

Fig. 4. Deformation microstructures of 4Al (a), 6Al (b, c), 8Al (d–f), and 10Al (g–i) steels after tensile tested. (a, b, d, e, g) Austenite. (c, f, h, i) Ferrite.

consequently leads to a significant multiplication of edge dislocations in 8Al steel.

Fig. 7(j–l) displays the dislocation structures at the ferrite-austenite phase interface. It can be observed that edge dislocations are more readily activated at the grain boundaries, which can be explained by the high stress present at these boundaries. It is reported that back stresses are generated in the soft phase, and forward stresses are developed in the hard phase during the deformation of heterogeneous structures [50]. In this case, the high stresses at ferrite grain boundaries, together with the stress-multiplication effect [51], facilitate the collective glide of dislocations on the same slip plane. Additionally, approximately 24.3 % austenite in 8Al steel, acting as a soft phase, can effectively alleviate stress localization within the ferrite and enhance the deformation of the ferrite to higher stresses. It is particularly interesting that edge, mixed, and screw dislocations in ferrite all exhibit a typical planar slip characteristic, which will be discussed in Section 4.2.

The above analysis reveals that dislocations in the ferrite phase of 10Al steel are predominantly identified as long straight screw dislocations, whose localized distribution contributes to brittle fracture behavior. In contrast, the ferrite of 8Al steel produces a significant number of edge dislocations and mixed dislocations with edge character, which markedly enhances the deformation capability of the 8Al steel ferrite.

4.2. *The planar slip of dislocations in ferrite*

In body-centered cubic (BCC) materials, screw dislocations typically exhibit pronounced cross-slip behavior due to their non-planar glide characteristics [52], which is fundamentally linked to the asymmetric core structure of screw dislocations and their high Peierls-Nabarro stress [53,54]. Interestingly, both edge and screw dislocations in 8Al steel and 10Al steel show distinct planar slip, as illustrated in Figs. 5 and 7. Studies demonstrate that the interaction between dislocations and ordered structures can lead to planar slip of dislocations, as observed in titanium alloys [30,31], FCC-structured Fe-Mn-Al-C alloys [12,13], and BCC-structured TiZrHfNbTa0.2 high-entropy alloys (HEAs) [32].

Fig. 8 shows the planar slip of dislocation in ferrite grains of 8Al steel and 10Al steel to reveal the deformation behaviors, which is controlled by the interaction between dislocation and the DO_3 ordered superlattice in the ferrite. It can be seen that the planar dislocation pattern can be

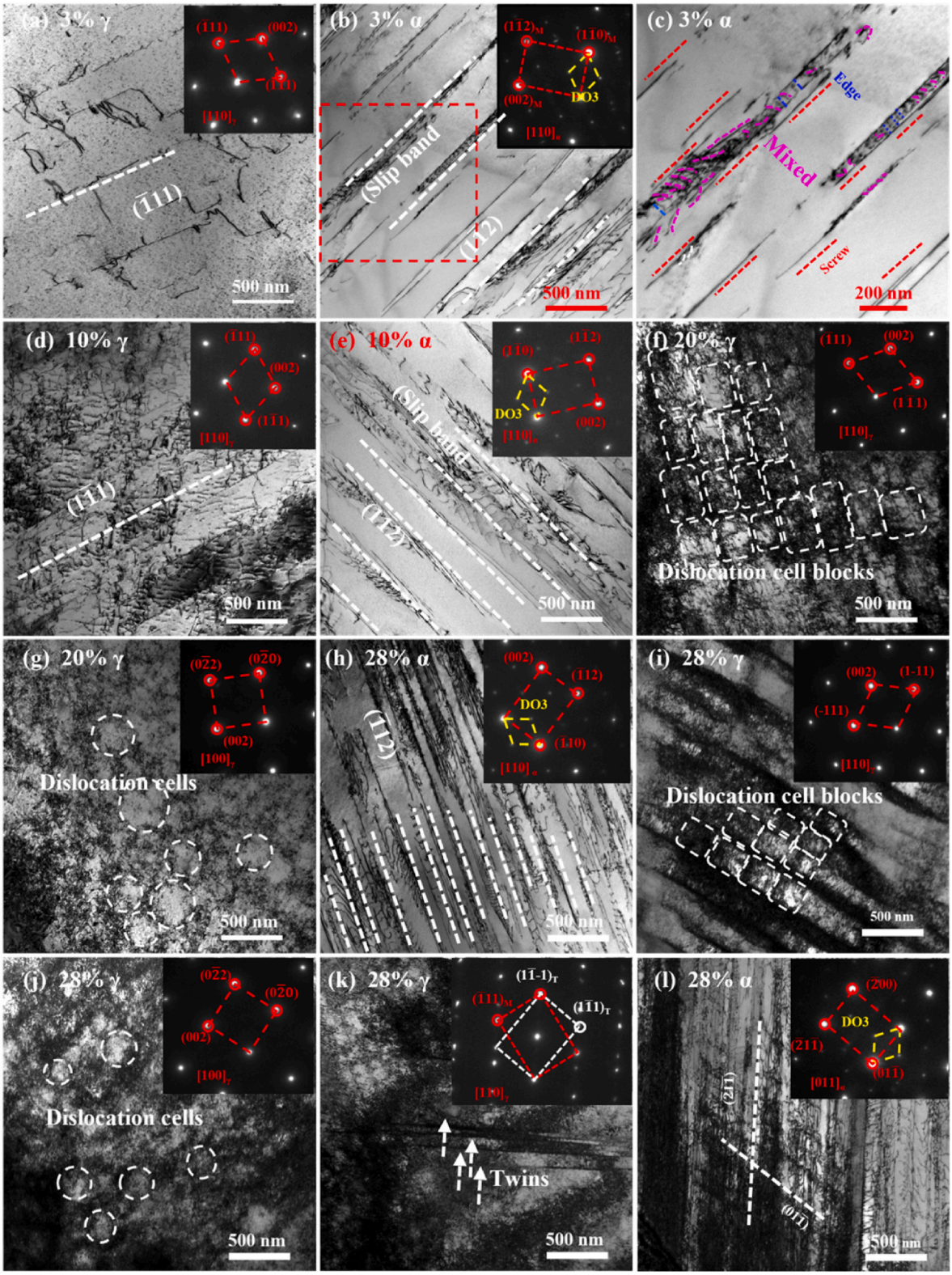

Fig. 5. Microstructures of austenite (a, d, f, g, i-k) and ferrite (b, c, e, h, l) in 8Al steel at different engineering strains. (a–c) 3 %, (d, e) 10 %, (f–h) 20 %, (i–l) 28 %.

easily observed in 8Al and 10Al steel, whose leading dislocations in the slip bands exhibit pairing behavior, as shown in Fig. 8(a, b, d, e). It is worth noting that in the 10Al steel, dislocation pairing is extremely pronounced, with up to six dislocation pairs observed and dislocation spacings as small as approximately 3 nm, demonstrating pronounced paired super-dislocation characteristics. It can be observed that the large sized DO_3 ordered domains in 10Al steel cannot be completely sheared (Fig. 8(c)), while the DO_3 ordered domains in 8Al steel can be completely sheared along the dislocation line direction and the slip plane direction (Fig. 8(f)). When the DO_3 ordered domains are destroyed by dislocations, anti-phase boundaries (APBs) are left behind [16]. The attractive force generated by APB energy leads to the pairing of

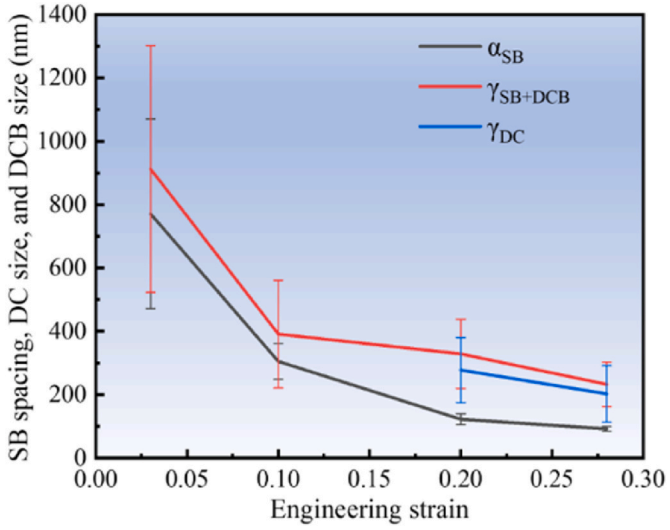

Fig. 6. The average slip band (SB) spacing, dislocation cell block (DCB) size, and dislocation cell (DC) size in austenite of the 8Al steel, and the average slip band spacing in ferrite of the 8Al steel at different deformation stages.

dislocations. Studies on Fe-Al alloys demonstrate that APB energy increases with Al content [14,55]. Consequently, the APB-induced attractive force in ferrite of 10Al steel becomes substantially stronger than that in ferrite of 8Al steel, resulting in more pronounced dislocation pairing in the 10Al alloy. Once the DO_3 ordered domain is completely sheared, no anti-phase boundaries (APBs) will be generated by subsequent dislocations [13]. The size of the superlattice ordered domains can be estimated by multiplying the number of paired dislocations by the Burgers vector [12].

$$D = Nb \tag{4}$$

where D is the size of superlattice ordered domains, N is the number of paired dislocations, and b = 0.25 nm [56] is the magnitude of the Burgers vector.

Fig. 8(b, e) shows that the number of dislocation pairs in 10Al steel and 8Al steel usually does not exceed 6 pairs and 3 pairs, respectively. With paired dislocations counts of 12 in 10Al steel and 6 in 8Al steel, the ordered domain sizes in 10Al and 8Al steels are estimated to be 3.0 nm and 1.5 nm, respectively. It is evident that the calculated DO_3 ordered domain size in ferrite of 10Al steel is significantly smaller than the TEM-observed value of 19 nm, whereas the calculated size for 8Al steel is very close to its TEM-observed value of 2.1 nm. Consequently, the ordered domains in ferrite of 8Al steel are more readily sheared, whereas the large-sized DO_3 ordered domains in ferrite of 10Al steel are difficult to be completely sheared.

To further illustrate the interaction between dislocations and the DO_3 ordered structure, inverse fast Fourier transforms (IFFT) are performed on the high-resolution images at positions 1 and 2 in Fig. 8(g) along the (01 $\bar{1}$) and (0 $\bar{1}$ 1) planes, as shown in Fig. 8(h and i). It can be observed that dislocations are composed of a multitude of dislocation dipoles, and the DO_3 superlattice is sheared along dislocation line and slip plane by a large number of these dislocation dipoles. These findings are in agreement with observations reported in Refs. [12,13,30,31] where sheared LRO clusters are proposed to cause planar slip.

4.3. Strain hardening behavior of 8Al steel

Fig. 3(d) reveals that 8Al steel exhibits multi-stage strain hardening, which is associated with the dynamic slip band refinement mechanism in ferrite. In stage A, the movement of dislocations leads to a sharp decrease in the strain hardening rate. In stage B, a reduction in the strain hardening rate is caused by the slip plane softening effect induced by dislocation shearing of the DO_3 long-range ordered domains [13]. In stage C, the passing stress generated by slip bands of stored dislocations has been demonstrated to be the primary contributor to strain hardening [12]. With the increase of strain, the number of slip bands increases, the spacing between slip bands decreases, and the passing stress rises, which leads to an increase in the strain hardening rate. In stage D, with the continued increase of strain, the slip bands become nearly saturated, and cross-slip of dislocations is activated, which leads to a decrease in the strain hardening rate. In stage E, localization of stress results in a rapid decline in the strain hardening rate and leads to plastic failure.

The deformation structures of ferrite and austenite in 8Al steel are dominated by dislocations until a small amount of deformation twins are activated in austenite near the maximum force. Therefore, the excellent strain hardening performance of 8Al steel is attributed to the refinement of ferrite slip bands and austenite dislocation cells. In the subsequent calculations, the flow stress of 8Al steel is estimated using the average spacing of ferrite slip bands and the average dislocation cell size of austenite. In dislocation-mediated plastic materials, the total flow stress σ_{tot} can be expressed by the following equation [57,58]:

$$\sigma_{tot} = \sigma_0 + \sigma(\varepsilon) \tag{5}$$

where σ_0 represents the deformation-independent strengthening contribution, i.e., the yield stress, and the strain hardening term $\sigma(\varepsilon)$ characterizes the evolution of flow stress during plastic deformation. The strain hardening term σ(ε) is composed of two parts, ferrite and austenite, and can be expressed as follows:

$$\sigma(\varepsilon) = 0.757\sigma_{\alpha}(\varepsilon) + 0.243\sigma_{\gamma}(\varepsilon) \tag{6}$$

where 75.7 % and 24.3 % represent the mass fractions of ferrite (α) and austenite (γ), respectively, and $\sigma_{\alpha}(\varepsilon)$ and $\sigma_{\gamma}(\varepsilon)$ denote the strain hardening terms for ferritic and austenitic phases during plastic deformation.

The ferrite strain hardening contribution from slip band refinement is determined through passing stress analysis [12,59]:

$$\sigma_{(\alpha)}(\varepsilon) = KMGb/D \tag{7}$$

where K is a geometrical factor, M = 2.8 [56,60] is the Taylor-factor, G = 80.7 GPa [56] is the shear modulus, b = 0.25 nm [56] is the magnitude of the Burgers vector, and D is the mean slip band spacing (Fig. 6).

The austenite strain hardening contribution arising from slip band, dislocation cell block, and dislocation cell refinement is derived from an empirical relationship [45,46,61]:

$$\sigma_{(\gamma)}(\varepsilon) = K_{SB}M_{SB}Gb/D + 0.8K_{DCB}M_{DCB}Gb/D + 0.2K_{DC}M_{DC}Gb/D \tag{8}$$

Where K is the similitude coefficient, M is the Taylor factor ($M_{SB} = M_{DCB}$ = 3.06, M_{DC} = 2.44) [46], G = 70 GPa [46] denotes the shear modulus, b = 0.25 nm [62] represents the magnitude of the Burgers vector, and D corresponds to the slip band spacing, dislocation cell block size, or dislocation cell size (Fig. 6). The values 0.8 and 0.2 are the area fractions of the dislocation cell blocks and the dislocation cells, respectively.

In order to calculate the coefficient of K in the austenite phase, Fig. 9 (a) presents the relation (8) for the 4Al steel, from which the similitude constant K is obtained as K_{SB+DCB} = 2.03, K_{DC} = 2.02. This value is very close to the reported K values of 2.0–2.9 for austenitic stainless steel [63, 64]. The austenite phases in 4Al and 8Al steels possess similar compositional systems and deformation structures. Studies [45,46] have shown that the similarity coefficient K essentially remains unchanged with variations in C and Al content for high manganese steels with similar deformation structures. Therefore, the similarity constant of austenite in 8Al steel is approximated by that in 4Al steel.

The calculated and measured (black line) results of the flow stress σ_{tot} for 8Al steel are shown in Table 4 and Fig. 9(b). The calculations are performed using Eqs. (5)–(8) and the data in Fig. 6. It can be seen that the best fit between the calculations and experiments is achieved when

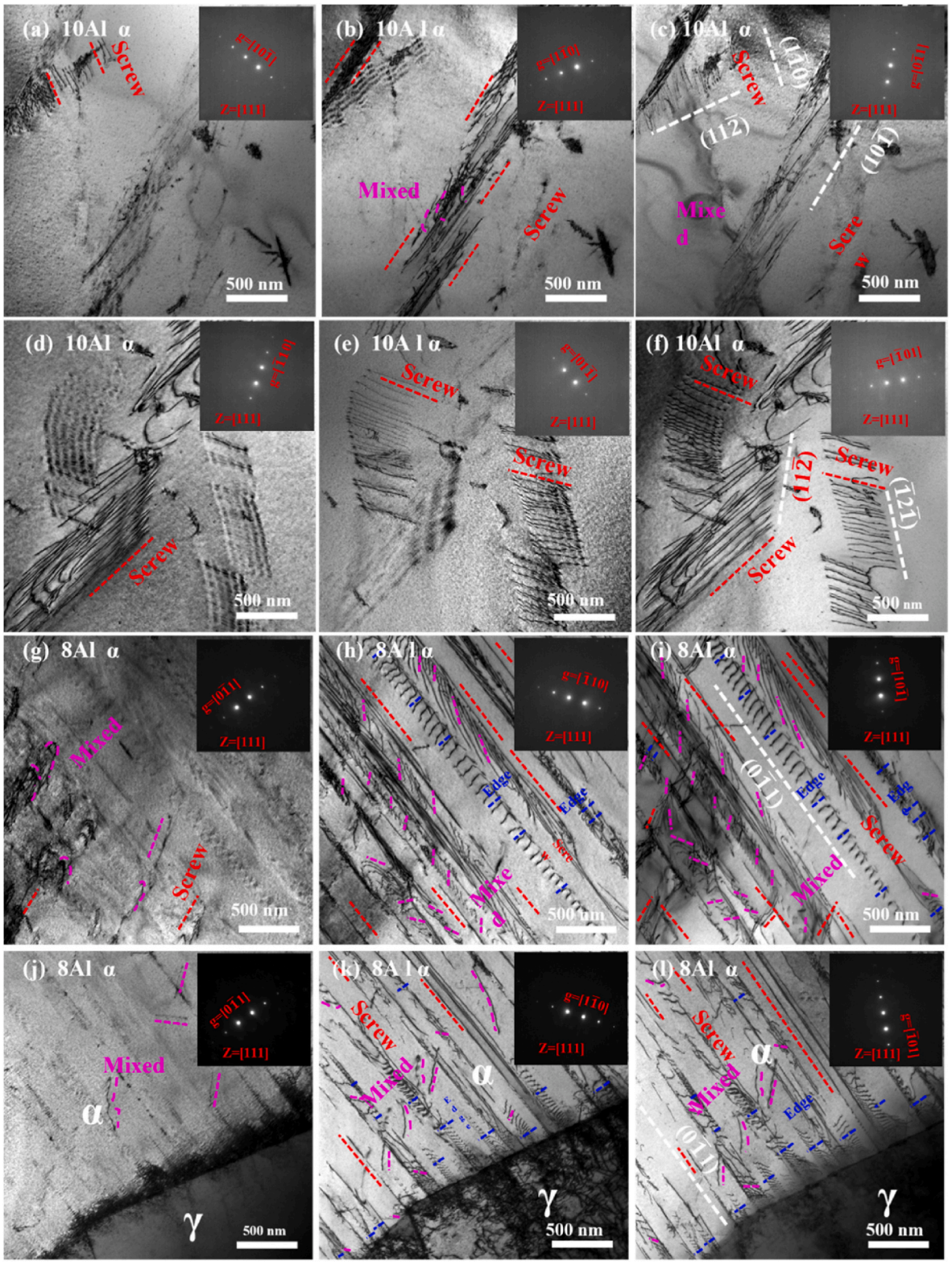

Fig. 7. Dislocation morphology of ferrite with different operation vectors after 3 % tensile deformation. (a)–(f) 10Al steel. (g)–(l) 8Al steel.

$K = 0.75$ and $\sigma_0 = 455$ MPa. This $K = 0.75$ value is very close to the similitude coefficient $K = 0.7$ for solid solution Fe30.4Mn8Al1.2C (wt%) austenite lightweight steel [12]. This further indicates that the ferrite slip bands have undergone the dynamic slip band refinement (DSBR) effect [11,12].

5. Conclusion

The initial microstructure, mechanical properties, and deformation microstructure evolution of Fe-35Mn-xAl-0.1C (x = 4, 6, 8, 10 wt%) low-carbon lightweight steels were investigated. This study not only reveals a new ductility-enhancing mechanism of ferrite + austenite duplex steel,

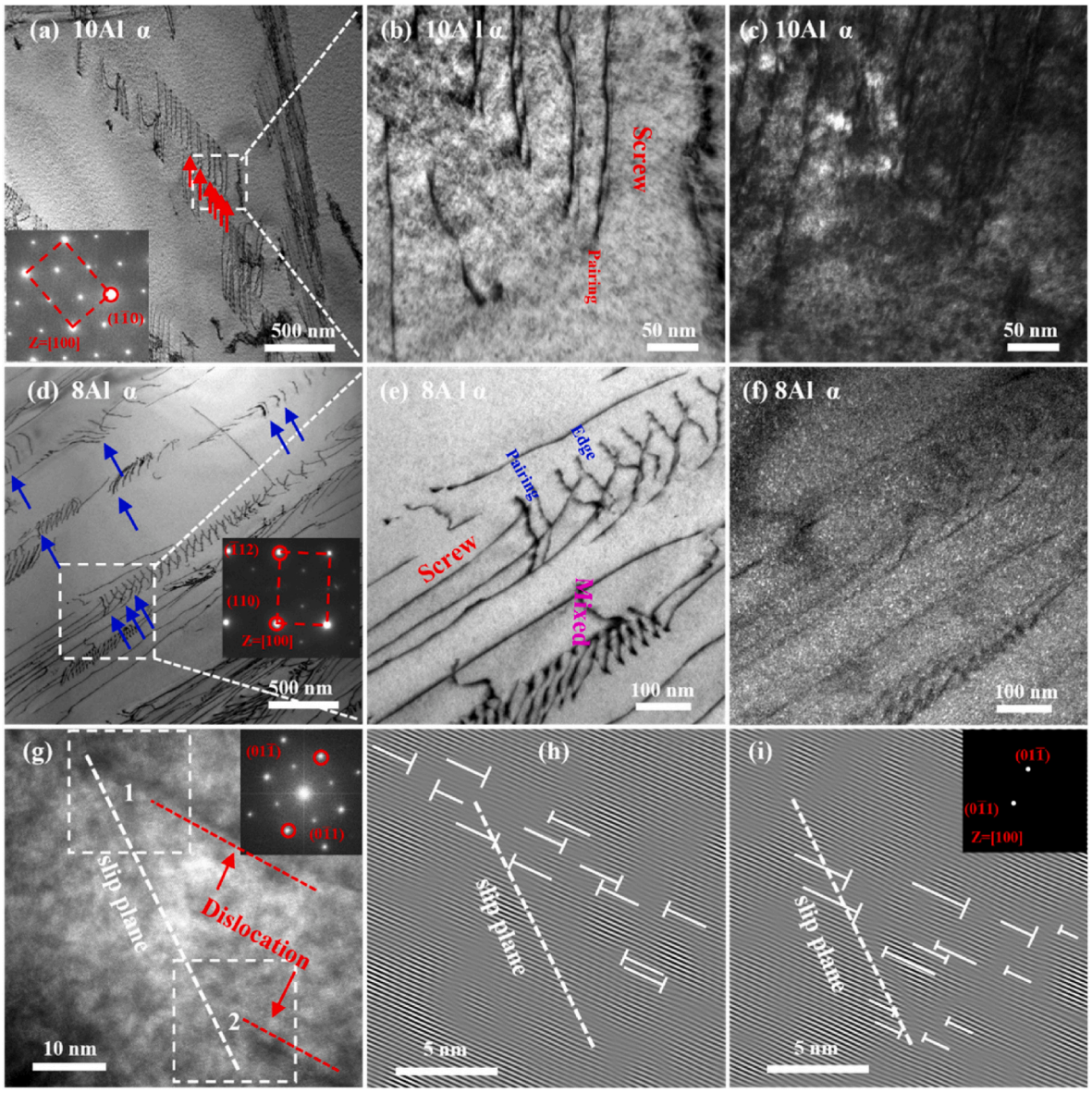

Fig. 8. The planar slip of dislocations in ferrite. (a–c) 10Al steel. (d–i) 8Al steel. (b, e) Enlarged view of a part of (a, d). (c, f) TEM dark field images using the (1-1-1) DO3 superlattice reflection. (g) High-resolution image along the [100] zone axis. (h, i) Inverse fast Fourier transform (IFFT) images of positions 1 and 2 in Fig. (g) along the (01 $\bar{1}$) and (0 $\bar{1}$ 1) planes, respectively.

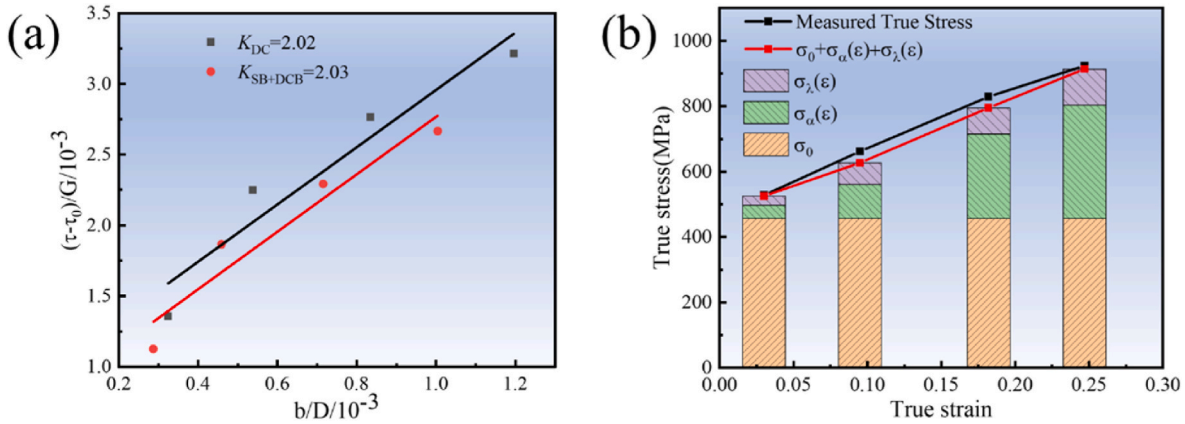

Fig. 9. Relationship between deformation microstructure and the flow stress. (a) Plot of the relation $(\tau-\tau 0)/G/10^{-3} = Kb/D/10^{-3}$ for the average slip band spacing, dislocation cell block size, and dislocation cell size of 4Al austenite steel, and (b) Calculated and measured stresses of the 8Al steel at different true strains.

Table 4
The strengthening contribution of $\sigma_{(\alpha)}(\varepsilon)$ and $\sigma_{(\gamma)}(\varepsilon)$ strengthening to the flow stress.

Engineering strain (%)	3.0	10.0	20.0	28.0
True strain, %	3.0	9.5	18.2	24.7
True stress, MPa	528	662	829	924
$\sigma_{(\alpha)}(\varepsilon)$, MPa	41	104	260	347
$\sigma_{(\gamma)}(\varepsilon)$, MPa	28	67	79	111

but also provides a new route for designing BCC based low-density steel with high strength and high ductility. Based on the experimental results, the following conclusions can be drawn:

1. With increasing Al content, the phase constitution transitions sequentially from single-phase austenite (4Al steel) to austenite-matrix duplex (6Al steel), then to ferrite-matrix duplex structures (8Al and 10Al steels). DO_3 ordered domains coexist within both the B2 ordered domains and the ferrite, exhibiting progressive size enlargement with higher Al concentrations.
2. The 8Al steel exhibits multi-stage strain hardening characteristics due to planar slip in ferrite. The dislocation structure in ferrite exhibits distinct planar slip, forming slip bands whose spacing decreases during straining, in accordance with the dynamic slip band refinement (DSBR) mechanism.
3. The ferritic dislocation structure in 10Al steel is predominantly characterized by localized long straight screw dislocations, which leads to plastic instability in 10Al steel, whereas in 8Al steel, it is characterized by a uniform distribution of long straight screw and edge dislocations. The high stress at the austenite-ferrite interface facilitates the activation of edge dislocations in ferrite.
4. The size of the shear zone is estimated based on the number of paired dislocations at slip band fronts. The estimated size in 8Al steel is less than 1.5 nm, which is very close to the TEM-observed value of less than 2.1 nm. The estimated size in 10Al steel is less than 3.0 nm, which is much smaller than the TEM-observed value of less than 19.0 nm.
5. It is found that as the Al content increases from 4 % to 8 %, the yield strength and tensile strength increase from 239 (±7) MPa and 552 (±12) MPa to 455 (±10) MPa and 74 (±14) MPa, respectively, while the uniform elongation only decreases from 38.5 (±2.0) % to 31.0 (±1.3) %. The synergistic operation of edge dislocation activity and DSBR effect in ferritic matrices underlies the superior strength-ductility balance observed in the 8Al ferrite-based duplex lightweight steel.

CRediT authorship contribution statement

Qingfeng Kang: Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Zexi Zhang:** Resources, Investigation. **Hui Wang:** Resources, Methodology, Investigation. **Haifeng Xu:** Formal analysis, Data curation. **Cunyu Wang:** Methodology, Investigation, Funding acquisition. **Zhengdong Liu:** Resources, Funding acquisition. **Wenquan Cao:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This research is supported by the Advanced Materials National Science and Technology Major Project (No. 2025ZD0611101) and Natural Nature Science Foundation of China (No. 51871062).

Data availability

Data will be made available on request.

References

[1] Q. Kang, Z. Zhang, Z. Gao, C. Wang, J. Liang, Z. Liu, Y. Weng, W. Cao, Excellent mechanical properties and underlined deformation twinning mechanism of a low-carbon low-density cryogenic steel, Mater. Sci. Eng., A 923 (2024), https://doi.org/10.1016/j.msea.2024.147741, 147741–147741.
[2] S. Chen, R. Rana, A. Haldar, R.K. Ray, Current state of Fe-Mn-Al-C low density steels, Prog. Mater. Sci. 89 (2017) 345–391, https://doi.org/10.1016/j.pmatsci.2017.05.002.
[3] Z. Gao, Q. Kang, H. Wang, H. Xu, J. Liang, W. Cao, Excellent strength and ductility synergy realized by hot forging and subsequent air cooling of a Fe-30Mn-11Al-1C-10Ni (wt.%) low-density steel, J. Mater. Sci. (2025), https://doi.org/10.1007/s10853-025-10655-6.
[4] Y.F. An, X.P. Chen, L. Mei, X.N. Tan, P. Ren, X.Y. Zhang, W.Q. Cao, Overcoming the strength-ductility trade-off dilemma in austenitic lightweight steel via stepwise controllable intragranular dual nanoprecipitation, Mater. Sci. Eng., A 908 (2024) 146773, https://doi.org/10.1016/j.msea.2024.146773.
[5] Z. Gao, Q. Kang, X. An, H. Wang, C. Wang, W. Cao, Enhanced mechanical properties of a Fe-Mn-Al-C austenitic low-density steel by increasing hot-rolling reduction, Mater. Char. 204 (2023) 113237, https://doi.org/10.1016/j.matchar.2023.113237.
[6] I. Gutierrez-Urrutia, A. Shibata, Effect of deformation temperature on strain localization phenomena in an austenitic Fe-30Mn-6.5Al-0.3C low-density steel, Acta Mater. (2023) 119566, https://doi.org/10.1016/j.actamat.2023.119566.
[7] Z. Zheng, H. Yang, A.P. Shatrava, J. Long, Y. Wang, J. Li, K. Zheng, Work hardening behavior and fracture mechanisms of Fe-18Mn-1.3C-2Cr low-density steel castings with varying proportions of aluminum alloying, Mater. Sci. Eng., A 862 (2023) 144467, https://doi.org/10.1016/j.msea.2022.144467.
[8] S. Hu, Z. Zheng, W. Yang, H. Yang, Fe–Mn–C–Al low-density steel for structural materials: a review of alloying, heat treatment, microstructure, and mechanical properties, Steel Res. Int. 93 (2022), https://doi.org/10.1002/srin.202200191.
[9] Y. Wang, Y. Zhang, A. Godfrey, J. Kang, Y. Peng, T. Wang, N. Hansen, X. Huang, Cryogenic toughness in a low-cost austenitic steel, Commun. Mater 2 (2021), https://doi.org/10.1038/s43246-021-00149-8.
[10] W. Cao, H. Wang, Z. Gao, C. Wang, J. Liang, A. Godfrey, L. Zhang, G. Wu, Synergetic strengthening from dynamic slip band-grain boundary interaction in a low-density femnalc steel, SSRN J. (2022), https://doi.org/10.2139/ssrn.4135342.
[11] H. Zhi, J. Li, W. Li, M. Elkot, S. Antonov, H. Zhang, M. Lai, Simultaneously enhancing strength-ductility synergy and strain hardenability via Si-alloying in medium-Al FeMnAlC lightweight steels, Acta Mater. 245 (2023) 118611, https://doi.org/10.1016/j.actamat.2022.118611.
[12] E. Welsch, D. Ponge, S.M. Hafez Haghighat, S. Sandlöbes, P. Choi, M. Herbig, S. Zaefferer, D. Raabe, Strain hardening by dynamic slip band refinement in a high-Mn lightweight steel, Acta Mater. 116 (2016) 188–199, https://doi.org/10.1016/j.actamat.2016.06.037.
[13] V. Gerold, H.P. Karnthaler, On the origin of planar slip in f.c.c. alloys, Acta Metall. 37 (1989) 2177–2183, https://doi.org/10.1016/0001-6160(89)90143-0.
[14] H.J. Leamy, F.X. Kayser, M.J. Marcinkowski, The plastic deformation behaviour of long-range ordered iron-aluminium alloys, Philos. Mag. 20 (1969) 763–777, https://doi.org/10.1080/14786436908228043.
[15] J. Herrmann, G. Inden, G. Sauthoff, Deformation behaviour of iron-rich iron-aluminum alloys at low temperatures, Acta Mater. 51 (2003) 2847–2857, https://doi.org/10.1016/s1359-6454(03)00089-2.
[16] D.G. Morris, M.A. Muñoz-Morris, L.M. Requejo, Work hardening in Fe–Al alloys, Mater. Sci. Eng., A 460–461 (2007) 163–173, https://doi.org/10.1016/j.msea.2007.01.014.
[17] X. Li, Z. Zhang, J. Wang, Deformation twinning in body-centered cubic metals and alloys, Prog. Mater. Sci. 139 (2023) 101160, https://doi.org/10.1016/j.pmatsci.2023.101160.
[18] S.I. Rao, C. Varvenne, C. Woodward, T.A. Parthasarathy, D. Miracle, O.N. Senkov, W.A. Curtin, Atomistic simulations of dislocations in a model BCC multicomponent concentrated solid solution alloy, Acta Mater. 125 (2017) 311–320, https://doi.org/10.1016/j.actamat.2016.12.011.
[19] D. Caillard, Kinetics of dislocations in pure Fe. Part I. In situ straining experiments at room temperature, Acta Mater. 58 (2010) 3493–3503, https://doi.org/10.1016/j.actamat.2010.02.023.
[20] Y.W. Wang, Q.F. He, Z.H. Ye, Y. Liao, C.W. Li, Q. Wang, J.F. Gu, Ductilization of single-phase refractory high-entropy alloys via activation of edge dislocation, Acta Mater. (2024), https://doi.org/10.1016/j.actamat.2024.120614, 120614–120614.
[21] L. Wang, J. Ding, S. Chen, K. Jin, Q. Zhang, J. Cui, B. Wang, B. Chen, T. Li, Y. Ren, S. Zheng, K. Ming, W. Lu, J. Hou, G. Sha, J. Liang, L. Wang, Y. Xue, E. Ma, Tailoring planar slip to achieve pure metal-like ductility in body-centred-cubic multi-principal element alloys, Nat. Mater. 22 (2023) 950–957, https://doi.org/10.1038/s41563-023-01517-0.
[22] P. Cizek, S.R. Kada, N. Armstrong, R.A. Antoniou, S. Slater, P.A. Lynch, Dislocation structures in a Ti–6Al–4V alloy subjected to cyclic tensile deformation, Mater. Sci. Eng., A 836 (2022) 142700, https://doi.org/10.1016/j.msea.2022.142700.

[23] F. Yang, R. Song, Y. Li, T. Sun, K. Wang, Tensile deformation of low density duplex Fe–Mn–Al–C steel, Mater. Des. 76 (2015) 32–39, https://doi.org/10.1016/j.matdes.2015.03.043.
[24] J.H. Hwang, T.T.T. Trang, O. Lee, G. Park, A. Zargaran, N.J. Kim, Improvement of strength – ductility balance of B2-strengthened lightweight steel, Acta Mater. 191 (2020) 1–12, https://doi.org/10.1016/j.actamat.2020.03.022.
[25] G. Frommeyer, U. Brüx, Microstructures and mechanical properties of high-strength Fe-Mn-Al-C light-weight TRIPLEX steels, Steel Res. Int. 77 (2006) 627–633, https://doi.org/10.1002/srin.200606440.
[26] G. Frommeyer, E.J. Drewes, B. Engl, Physical and mechanical properties of iron-aluminium-(Mn, Si) lightweight steels, Rev. Met. Paris 97 (2000) 1245–1253, https://doi.org/10.1051/metal:2000110.
[27] S.-H. Kim, H. Kim, N.J. Kim, Brittle intermetallic compound makes ultrastrong low-density steel with large ductility, Nature 518 (2015) 77–79, https://doi.org/10.1038/nature14144.
[28] N.S. Stoloff, Iron aluminides: present status and future prospects, Mater. Sci. Eng., A 258 (1998) 1–14, https://doi.org/10.1016/s0921-5093(98)00909-5.
[29] M.J. Marcinkowski, M.E. Taylor, F.X. Kayser, Relationship between atomic ordering and fracture in Fe-Al alloys, J. Mater. Sci. 10 (1975) 406–414, https://doi.org/10.1007/bf00543684.
[30] T. Neeraj, D.-H. Hou, G.S. Daehn, M.J. Mills, Phenomenological and microstructural analysis of room temperature creep in titanium alloys, Acta Mater. 48 (2000) 1225–1238, https://doi.org/10.1016/s1359-6454(99)00426-7.
[31] R. Zhang, S. Zhao, C. Ophus, Y. Deng, S.J. Vachhani, B. Ozdol, R. Traylor, K. C. Bustillo, J.W. Morris, D.C. Chrzan, M. Asta, A.M. Minor, Direct imaging of short-range order and its impact on deformation in Ti-6Al, Sci. Adv. 5 (2019), https://doi.org/10.1126/sciadv.aax2799.
[32] X. Wen, L. Zhu, M. Naeem, H. Huang, S. Jiang, H. Wang, X. Liu, X. Zhang, X.-L. Wang, Y. Wu, Z. Lu, Strong work-hardenable body-centered-cubic high-entropy alloys at cryogenic temperature, Scr. Mater. 231 (2023) 115434, https://doi.org/10.1016/j.scriptamat.2023.115434.
[33] S. Curtze, V.-T. Kuokkala, Dependence of tensile deformation behavior of TWIP steels on stacking fault energy, temperature and strain rate, Acta Mater. 58 (2010) 5129–5141, https://doi.org/10.1016/j.actamat.2010.05.049.
[34] S. Curtze, V.-T. Kuokkala, A. Oikari, J. Talonen, H. Hänninen, Thermodynamic modeling of the stacking fault energy of austenitic steels, Acta Mater. 59 (2011) 1068–1076, https://doi.org/10.1016/j.actamat.2010.10.037.
[35] J.D. Yoo, K.-T. Park, Microband-induced plasticity in a high Mn–Al–C light steel, Mater. Sci. Eng. A (2008) 496. https://kns.cnki.net/kcms/detail/detail.aspx?dbname=garj0010_4&filename=sjes4cab9e082a885c1e15a15751dd2b8bc1.
[36] W.K. Choo, J.H. Kim, J.C. Yoon, Microstructural change in austenitic Fe-30.0wt% Mn-7.8wt%Al-1.3wt%C initiated by spinodal decomposition and its influence on mechanical properties, Acta Mater. 45 (1997) 4877–4885, https://doi.org/10.1016/s1359-6454(97)00201-2.
[37] K.H. Yang, W.K. Choo, Evidence of carbon ordering and morphology change in a cubic carbide phase, Philos. Mag. Lett. 62 (1990) 221–226, https://doi.org/10.1080/09500839008215062.
[38] W.B. Lee, F.R. Chen, S.K. Chen Z, G.B. Olson, C.M. Wan, TRANSMISSION ELECTRON MICROSCOPY STUDIES OF THE CRYSTALLOGRAPHY OF B.C.C./18R MARTENSITE IN Fe-Mn-A1-C, (n.d.).
[39] A. Kazaryan, Y. Wang, S.A. Dregia, B.R. Patton, Grain growth in anisotropic systems: comparison of effects of energy and mobility, Acta Mater. 50 (2002) 2491–2502, https://doi.org/10.1016/s1359-6454(02)00078-2.
[40] A. Kazaryan, B.R. Patton, S.A. Dregia, Y. Wang, On the theory of grain growth in systems with anisotropic boundary mobility, Acta Mater. 50 (2002) 499–510, https://doi.org/10.1016/s1359-6454(01)00369-x.
[41] V.D.F.C. Lins, M. Afonso Freitas, E. Mirra De Paula e Silva, Oxidation kinetics of an Fe–31.8Mn–6.09Al–1.60Si–0.40C alloy at temperatures from 600 to 900 °C, Corros. Sci. 46 (2004) 1895–1907, https://doi.org/10.1016/j.corsci.2003.10.015.
[42] L. Zhang, R. Song, C. Zhao, F. Yang, Y. Xu, S. Peng, Evolution of the microstructure and mechanical properties of an austenite–ferrite Fe–Mn–Al–C steel, Mater. Sci. Eng., A 643 (2015) 183–193, https://doi.org/10.1016/j.msea.2015.07.043.
[43] L. Xu, H. Wu, Microstructural evolution and mechanical property optimization under solution treatment of an ultra-low carbon Fe-Mn-Al duplex steel, Mater. Sci. Eng., A 738 (2018) 163–173, https://doi.org/10.1016/j.msea.2018.09.097.
[44] M. Liu, C. Song, Z. Cui, Crystallographic texture evolution and martensite transformation in the strain hardening process of a ferrite-based low density steel, J. Mater. Sci. Technol. 78 (2021) 247–259, https://doi.org/10.1016/j.jmst.2020.11.034.
[45] I. Gutierrez-Urrutia, D. Raabe, Dislocation and twin substructure evolution during strain hardening of an Fe–22wt.% Mn–0.6wt.% C TWIP steel observed by electron channeling contrast imaging, Acta Mater. 59 (2011) 6449–6462, https://doi.org/10.1016/j.actamat.2011.07.009.
[46] I. Gutierrez-Urrutia, D. Raabe, Multistage strain hardening through dislocation substructure and twinning in a high strength and ductile weight-reduced Fe–Mn–Al–C steel, Acta Mater. 60 (2012) 5791–5802, https://doi.org/10.1016/j.actamat.2012.07.018.
[47] X. Ma, J. Chen, Y. Liu, X. Wang, S. Huang, Z. Chen, Effect of short-range order on microstructure, texture and strain hardening of cold drawn Cu-10 at.%Mn alloy, Mater. Char. 135 (2018) 32–39, https://doi.org/10.1016/j.matchar.2017.11.024.
[48] M. Yamaguchi, Y. Umakoshi, The deformation behaviour of intermetallic superlattice compounds, Prog. Mater. Sci. 34 (1990) 1–148, https://doi.org/10.1016/0079-6425(90)90002-q.
[49] F. Wang, G.H. Balbus, S. Xu, Y. Su, J. Shin, P.F. Rottmann, K.E. Knipling, J.-C. Stinville, L.H. Mills, O.N. Senkov, I.J. Beyerlein, T.M. Pollock, D.S. Gianola, Multiplicity of dislocation pathways in a refractory multiprincipal element alloy, Science 370 (2020) 95–101, https://doi.org/10.1126/science.aba3722.
[50] S. Liu, G. Huang, M. Gupta, B. Jiang, F. Pan, Microstructures, mechanical properties and deformation mechanism of heterogeneous metal materials: a review, J. Mater. Sci. Technol. (2025), https://doi.org/10.1016/j.jmst.2025.03.109.
[51] M. Jouiad, N. Cleíment, A. Coujou, Friction Stresses in the G Phase of a Nickel-based Superalloy, (n.d.).
[52] C.R. Weinberger, B.L. Boyce, C.C. Battaile, Slip planes in bcc transition metals, Int. Mater. Rev. 58 (2013) 296–314, https://doi.org/10.1179/1743280412y.0000000015.
[53] V. Vitek †, Core structure of screw dislocations in body-centred cubic metals: relation to symmetry and interatomic bonding, Philos. Mag. 84 (2004) 415–428, https://doi.org/10.1080/14786430310001611644.
[54] M.S. Duesbery, V. Vitek, Plastic anisotropy in b.c.c. transition metals, Acta Mater. 46 (1998) 1481–1492, https://doi.org/10.1016/s1359-6454(97)00367-4.
[55] H.J. Lea, F.X. Kayser, The Compressive Deformation Behavior of Long Range Ordered Polycrystalline Iron-Aluminum Alloys'), (n.d.).
[56] T. Liu, Z. Cao, H. Wang, G. Wu, J. Jin, W. Cao, A new 2.4 GPa extra-high strength steel with good ductility and high toughness designed by synergistic strengthening of nano-particles and high-density dislocations, Scr. Mater. 178 (2019) 285–289, https://doi.org/10.1016/j.scriptamat.2019.11.045.
[57] U.F. Kocks, H. Mecking, Physics and phenomenology of strain hardening: the FCC case, Prog. Mater. Sci. 48 (2002) 171–273, https://doi.org/10.1016/s0079-6425(02)00003-8.
[58] D. Kuhlmann-Wilsdorf, H.G.F. Wilsdorf, J.A. Wert, LEDS theory of workhardening stages and “planar” versus “distributed” glide, Scripta Metall. Mater. 31 (1994) 729–734, https://doi.org/10.1016/0956-716x(94)90218-6.
[59] H. Neuhauser, J. Koropp, R. Heege, Electron microscopic studies in the yield region of 70/30-α-brass single crystals—I. The mode of slip, Acta Metall. 23 (1975) 441–453, https://doi.org/10.1016/0001-6160(75)90083-8.
[60] J.-S. Wang, M.D. Mulholland, G.B. Olson, D.N. Seidman, Prediction of the yield strength of a secondary-hardening steel, Acta Mater. 61 (2013) 4939–4952, https://doi.org/10.1016/j.actamat.2013.04.052.
[61] D. Kuhlmann-Wilsdorf, Theory of plastic deformation: - properties of low energy dislocation structures, Mater. Sci. Eng., A 113 (1989) 1–41, https://doi.org/10.1016/0921-5093(89)90290-6.
[62] O. Bouaziz, S. Allain, C. Scott, Effect of grain and twin boundaries on the hardening mechanisms of twinning-induced plasticity steels, Scr. Mater. 58 (2008) 484–487, https://doi.org/10.1016/j.scriptamat.2007.10.050.
[63] X. Feaugas, ON THE ORIGIN OF THE TENSILE FLOW STRESS IN THE STAINLESS STEEL AISI 316L AT 300 K: BACK STRESS AND EFFECTIVE STRESS, (n.d.).
[64] M. Sauzay, L.P. Kubin, Scaling laws for dislocation microstructures in monotonic and cyclic deformation of fcc metals, Prog. Mater. Sci. 56 (2011) 725–784, https://doi.org/10.1016/j.pmatsci.2011.01.006.